

Project Scope

Project Background

The current powder transport and packing systems require operator intervention during plant shutdown and packer empty-out activities. Existing PLC logic partially supports these functions; however, several sequences remain incomplete, unimplemented, or require manual intervention to fully empty transport lines, storage bins, and vacuum conveying systems.

This project aims to improve automation efficiency, reduce operator intervention, minimize residual powder hold-up, and standardize shutdown and empty-out processes through PLC programming modifications and potential hardware enhancements.

Project Objectives

1. Fully automate transport system purge during plant shutdown.
 2. Prevent packer empty-out sequences from commencing until transport purge is complete.
 3. Automate packer empty-out processes currently requiring manual intervention.
 4. Improve powder recovery from storage bins and vacuum conveying systems.
 5. Minimize residual powder remaining in bins, transport lines, and vacuum cans.
 6. Provide configurable operating parameters where appropriate.
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Scope of Work

1. Transport System Shutdown Purge Sequence

Existing Issues

- An automated transport line purge exists but activation logic requires verification.
- Current shutdown behaviour may not completely empty transport lines or alternate destination bins.
- Selection **PwdPRWP12 – Drain Tipping Transport Line** exists but contains no implemented PLC logic.

Proposed Modifications

Shutdown Sequence

Modify PLC logic to:

1. Allow normal transport operation until:
 - Sifter shutdown complete.
 - Fluid bed shutdown complete.

1. Initiate transport line purge:
 - Purge initially to the currently selected destination bin.
 - Automatically switch and purge to the alternate bin.
2. Upon completion of transport purge:
 - Trigger one additional vacuum can cycle.
 - Verify all low-level and high-level powder conditions are cleared.
 - Return system to standby/shutdown state.

Interlocking

Implement interlock preventing packer empty-out sequence from starting until:

- Transport purge sequence has completed successfully.

Investigation Required

- Determine purpose and intended functionality of:
 - **PwdPRWP12 – Drain Tipping Transport Line**

Deliverables

- Updated PLC shutdown sequence.
 - Updated HMI status indications.
 - Verification testing during shutdown operation.
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2. Packing Empty-Out Automation

Existing Situation

Current empty-out process requires operator intervention.

Investigation Required

Document and review:

- Current operator actions.
- Existing SOPs.
- Manual steps performed by packing operators.

Proposed Modifications

Develop automated PLC sequence to:

- Replicate current operator empty-out procedure.
- Minimize manual intervention.
- Ensure consistent packer emptying process.

Deliverables

- Functional design update.
 - PLC sequence development.
 - Operational testing and validation.
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3. Bin-to-Packer Empty-Out Improvements

Existing Issues

Current empty-out sequence performs adequately but:

- Bins 1 and 3 feeding A Packer retain approximately:
 - 50 kg to 100 kg of powder.
- Issue caused by:
 - Pipe arrangement.
 - Shape of new bustle design.

Potential Hardware Modification

Investigate installation of bin vibrators on:

Powder Silo 1 to Bulk Packer

- PwdS1P27
- PwdS1P28

Powder Silo 3 to Bulk Packer

- PwdS3P27
- PwdS3P28

PLC Modifications

If vibrators are installed:

- Enable automatically during transfer steps.
- Disable once transfer/empty-out sequence completes.
- Include alarm and permissive handling.

Additional Requirement

Empty-out sequence must automatically empty:

- Both bins aligned to plant operation.

Currently this activity is performed manually by operators.

Investigation Required

Obtain:

- Existing operator procedure.
- Detailed empty-out methodology.
- SOP documentation.

Deliverables

- PLC modifications.
 - Vibrator control integration (if approved).
 - Functional testing.
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4. Purge Time Enhancement

Requirement

Review and modify purge timing strategy to:

- Extend purge duration where required.
- Allow configurable purge time override/setpoint.

Deliverables

- Configurable PLC parameters.
 - HMI operator settings page.
 - Documentation update.
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5. Vacuum Can Empty-Out Automation

Existing Situation

Operators manually empty the four vacuum cans after bin empty-out timer completion.

Proposed Automation

Automate final can emptying process by:

- Executing two additional vacuum conveying cycles after bin empty-out timer completes.

Applicable Equipment

Packer A Vacuum Cans

- PwdBlkAP11
- PwdBlkAP12
- PwdBlkAP13
- PwdBlkAP14

Reference:

- FD Bulk Packer A.xlsm

PLC Logic Modification

Repeat the following sequence steps:

- Step 16
- Step 18
- Step 20

for configurable repeat cycles.

Recommended Enhancement

Create adjustable parameter:

Additional Empty-Out Cycles

- Default = 2 cycles
- Adjustable by authorized personnel

Deliverables

- PLC logic enhancement.
- Operator-adjustable cycle parameter.
- Testing and commissioning.

Exclusions

The following items are excluded unless separately approved:

- Mechanical installation of vibrators.
- Structural modifications to silos, bins, pipework or bustles.
- Replacement of conveying equipment.
- Major redesign of transport routing.
- Third-party OEM software modifications outside PLC control scope.

Assumptions

1. Existing PLC hardware has sufficient capacity for logic additions.
2. Existing HMI system can accommodate required screens and parameters.
3. Current functional descriptions and PLC code accurately reflect plant operation.
4. Operations personnel will provide SOPs describing current manual processes.

5. Access to Tetra Pak/Macro support will be available if required.

Key Risks

Risk	Impact
Existing process knowledge incomplete	Additional engineering effort
Lack of documented manual procedures	Delays to automation design
Residual powder caused by mechanical design limitations	May require hardware solutions
Undefined PwdPRWP12 functionality	Potential scope growth
Production testing availability	Commissioning delays

Success Criteria

- Transport lines fully purged during plant shutdown.
- Packer empty-out interlocked with transport purge completion.
- Reduction in manual operator intervention.
- Automated emptying of vacuum cans.
- Reduced powder hold-up in bins and conveying systems.
- Configurable purge and empty-out parameters available through HMI.
- Successful commissioning and operator sign-off.